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Review article

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Optimizing the delivery of radiotherapy with artificial intelligence

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Abstract

Artificial intelligence (AI) and machine learning are transformative technologies that have sparked both excitement and concern. In radiation oncology, AI has been successfully applied to automate tasks, such as auto-contouring, geometric adaptation, calculation of radiation dose and quality assurance, and predicting outcomes to provide optimal patient care. The clinical deployment of AI-driven and machine learning-based tools, however, continues to lag behind their perceived potentials, owing to a range of technical, practical, ethical and legal concerns. The original predictive AI algorithms have been expanded with technologies such as generative AI and foundation models to enable new applications, such as automated treatment planning and synthetic computed tomography generation, as well as improved prediction of individual patient outcomes. The increasing availability of innovations such as agentic AI, digital twins and multimodal AI could further improve the delivery and, thus, the efficacy of radiotherapy. In this Review, we present examples of successful AI applications in radiation oncology, provide an overview of the current challenges for implementing such tools in this specialty and strategies for addressing them, and discuss the broader implications of these tools in optimizing treatment outcomes and the quality of patient care.

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Key Points

- Artificial intelligence (AI) is currently transforming radiation oncology workflows and patient care through a growing number of commercially available and Food and Drug Administration-approved tools.
- AI is transforming the defined roles of radiation oncology care providers by automating routine tasks and supporting predictive and clinical decision-making, while expert human oversight remains essential across AI-enabled workflows.
- Many ongoing clinical trials are evaluating the technical performance and feasibility of AI-enabled applications for radiation oncology, and clinical research is now starting to address the safety, efficacy and clinical value of these tools.
- Despite substantial progress, several challenges hinder the effective implementation of AI tools into routine radiation oncology workflows, and the field needs practical recommendations to support responsible clinical adoption of these tools.
- The convergence of digital twins, generative AI and foundation models is poised to shape the next era of AI-enabled radiotherapy.

Introduction

The practice of radiation oncology involves complex, data-intensive clinical workflows encompassing heterogeneous sources of information at every stage — from initial consultation and imaging to treatment planning, quality assurance and delivery of radiotherapy. The aim is to ensure that patients receive the prescribed radiation treatment safely and accurately. These steps can vary and involve several stages of sophisticated human–machine interactions and decision-making, such as chart review, machine-specific and patient-related quality assurance, tumour and/or organ contouring, treatment planning, image guidance, management of respiratory motion, response modelling and prediction of outcomes[1]. The body of experience in this field grows with every newly treated patient, creating a wealth of information, which, combined with the error-averse nature of radiation oncology, makes this medical specialty an ideal setting for leveraging the unique capabilities of artificial intelligence (AI)[2–4].

In this Review, we present examples of successful AI applications in radiation oncology. We further delve into the current challenges of successful clinical implementation of AI, machine learning and digital twins in this specialty by outlining key requirements and exploring strategies that integrate scientific discovery with engineering to address these obstacles, including the need for training programmes to enable health-care professionals to meet the requirements of the AI era. We also discuss the broader implications of using AI tools in radiation oncology, such as the need to ensure safe and optimal use of AI and machine learning in order to optimize the quality of care and, ultimately, patient outcomes.

AI applications in radiation oncology

Key AI-related concepts in radiation oncology

The use of AI in radiation oncology has its roots in the mid-1990s[5]. The first applications mainly focused on predictive AI technologies, which typically forecast optimal treatment plans or predict clinical outcomes, such as treatment response or adverse events, on the basis of provided data and have been reviewed elsewhere[2]. These tools involved training artificial neural networks for automating the evaluation of treatment planning[6], customizing beam orientation[7] or standardizing treatment plans (an approach referred to as knowledge-based planning (KBP))[8], to name a few examples. Later applications developed in the mid-2000s focused on predicting treatment outcomes across cancer sites[9–13]. With the resurgence of deep learning methods[5], many applications repurposed such methods for auto-contouring[14,15], dose prediction[16–18], treatment planning[19], outcome modelling using imaging (radiomics)[20] or genetics (radiogenomics)[21], and optimizing decision-making[22]. Predictive AI methodologies in radiation oncology have traditionally leveraged statistical learning and machine learning approaches, such as logistic regression, gradient boosting frameworks, support vector machines and deep learning classifiers[5]. These models are primarily designed to forecast clinical end points, including response to treatment, survival and disease progression, as well as to automate workflow tasks such as organ segmentation and optimization of treatment planning (Fig. 1).

Subsequent advances have focused on the application of generative AI technologies, which are able to produce new content (whether that be text, images or videos) from given inputs[23]. Generative AI encompasses models capable of synthesizing data and includes large language models (LLMs) and vision–language models (VLMs), generative adversarial networks (GANs), diffusion models, variational autoencoders and transformer-based generative frameworks[24]. Radiation oncology is currently leveraging a combination of both predictive and generative AI capabilities. These methods have complementary strengths: predictive models support automated treatment planning, stratification on the basis of toxicity risk and forecasting of response to treatment[25,26], whereas generative AI facilitates data augmentation, feature representation and synthesis of multimodal clinical information to strengthen predictive model robustness and clinical decision-making. In addition, the AI technologies applied in radiation oncology can be divided in terms of scale into narrow and general AI, which differ in the sense that narrow AI is specialized for a single task whereas general AI models (referred to as foundation models) can be adapted to multiple tasks. Foundation models are typically large-scale AI models that are trained on a vast amount of often unlabelled, diverse data using techniques such as self-supervised and/or weak-supervised learning, enabling their adjustment for many different tasks. Multimodal foundation models, including VLMs that jointly process imaging and textual information, enable integration of radiological images, treatment planning data and clinical text. Generative AI models can produce hallucinations, which constitute an important safety concern in clinical applications[24].

Agentic AI and digital twins

The emergence of digital twins is a key companion development to AI. The US National Academies of Sciences, Engineering, and Medicine has formally defined digital twins as "a set of virtual information constructs that mimics the structure, context and behaviour of a natural, engineered or social system (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value"[27]. In this symbiotic relationship[28], AI provides digital twins with computing cognition (a 'brain') whereas digital twins provide AI with an engineered context (an 'environment').

Agentic AI builds on generative AI and LLMs not only by producing content such as structured clinical notes, treatment summaries or protocol recommendations, but also by using that information to carry

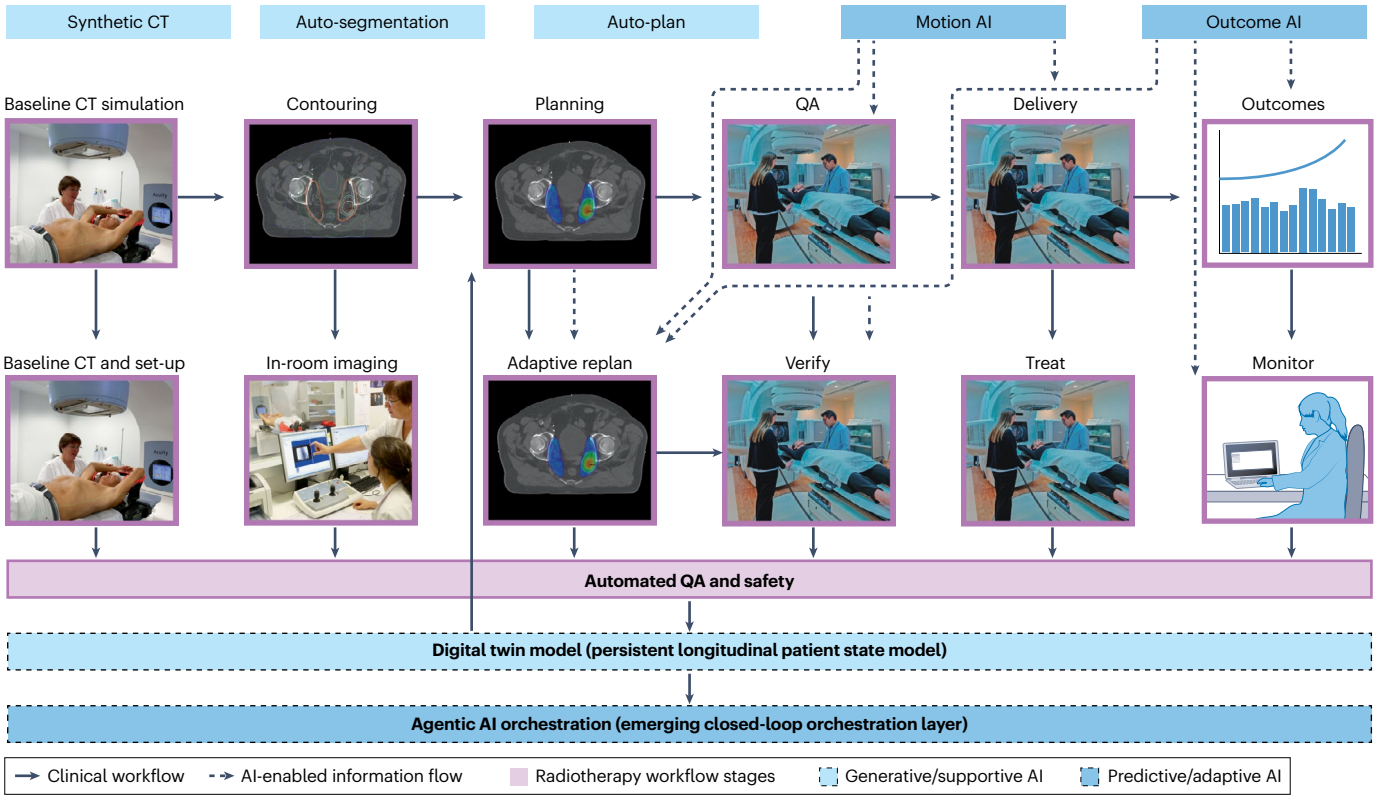


Fig. 1 | Integration of AI, digital twins and agentic AI orchestration across the radiation oncology workflow. This conceptual framework represents three interconnected components: the conventional radiotherapy workflow, artificial intelligence (AI) applications integrated across workflow stages, and digital twin and agentic AI infrastructure layers that enable longitudinal monitoring, coordination and adaptive decision support. The upper workflow illustrates standard radiotherapy processes, including simulation, contouring, treatment planning, quality assurance (QA), treatment delivery and longitudinal monitoring of outcomes. Generative and supportive AI applications that can be used in these processes include generation of synthetic computed tomography (CT) images, automated segmentation and treatment-planning assistance, whereas predictive and adaptive AI systems can be applied in motion assessment, treatment adaptation and outcome prediction. AI-enabled information pathways include predictive guidance, adaptive recommendations and longitudinal data exchange between workflow stages, rather than direct autonomous control. As indicated through vertical connections with the QA layer, automated verification and safety monitoring are performed continuously throughout the treatment process using automated QA and safety systems integrated throughout the workflow. The digital twin framework is a persistent longitudinal patient-state model integrating imaging, clinical, treatment and outcome data over time. Finally, an emerging agentic AI orchestration framework is being designed to coordinate communication across AI systems and support adaptive workflows, and might enable future closed-loop decision support across the radiotherapy continuum. Credits: Thomas Hecker/Alamy (Contouring, Planning and Adaptive replan); Album/Album/Alamy (Baseline CT simulation, Baseline CT and set-up, and In-room imaging); Rob Walls/Alamy (QA, Delivery, Verify, Treat).

out multistep tasks across radiation oncology facilities. An agent can extract data from the electronic health record, monitor and predict treatment-related adverse events over time, identify and flag patients at risk of treatment complications, and coordinate actions to follow up with these patients, without manual prompting at every step[29,30]. An illustrative example is GPT-plan, a multi-agent LLM-based system that mimics the clinical workflow of dosimetrists and physicists to iteratively generate and evaluate radiotherapy plans, achieving high-quality plans with reduced optimization iterations[31].

Despite advances in agentic AI applications in oncology, clinical validation remains limited, with only one randomized controlled trial reported thus far to the best of our knowledge[32]. This trial tested a lifestyle intervention involving personalized physical activity coaching to increase daily step counts, incorporating autonomous tracking using a wearable sensor and goal-directed behavioural optimization through multimodal interaction (for example, with electronic tablets and/or smart glasses), and multi-agent collaborations. Most studies in this space, however, rely on simulated data and lack validation in real-world practice, highlighting the fact that agentic AI is still in the early stages of development.

In radiation oncology, digital twins are currently being used together with predictive AI to improve workflow efficiency[33] and will also constitute the natural companion of agentic AI by providing a dynamic, patient-specific representation of anatomy, treatment delivery and longitudinal outcomes. The bidirectional interaction between the physical patient and its digital counterpart is central to this concept, and it enables the AI agents to iteratively learn and adapt to the task at hand. This combined framework positions digital twins–agentic AI as a powerful paradigm pairing for personalized medicine. Although fully autonomous digital twin ecosystems remain investigational, early digital twin frameworks have already demonstrated feasibility in adaptive radiotherapy and for modelling response to treatment[33–35].

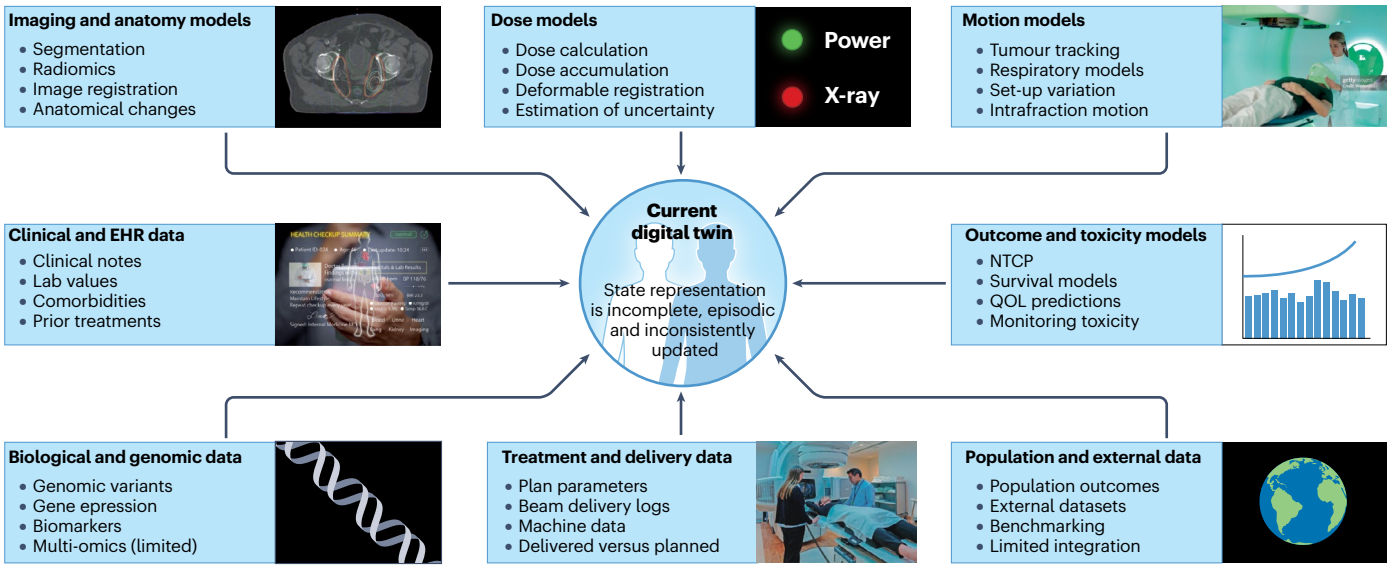


Fig. 2 | Current implementation of digital twin models in radiation oncology. Digital twin frameworks comprise multiple task-specific models spanning imaging and anatomy, dose calculation, motion assessment, outcome prediction, clinical data, biological data and treatment-delivery information. These individual components provide valuable insights and are often developed and deployed in isolation, with limited interoperability and inconsistent longitudinal integration. EHR, electronic health record; NTCP, non-malignant tissue complication probability; QOL, quality of life. Credits: Thomas Hecker/Alamy (Imaging and anatomy models); Westend61/Westend61/Getty (Motion models); Rob Walls/Alamy (Treatment and delivery data); Tanapong Sungkaew/Alamy (Clinical and EHR data).

Instead of relying on static datasets or single timepoint predictions, digital twins continuously integrate multimodal AI data, including imaging, electronic health records, dosimetry, biomarkers and patient-reported outcomes (PROs), to simulate response to and toxicities from treatment over time[36,37]. Early implementations of digital twin models relevant to radiation oncology have demonstrated feasibility in generating patient-specific predictive models of response to radiotherapy (Fig. 2). For example, digital twin frameworks integrating tumour growth dynamics with magnetic resonance imaging (MRI) data have been used to optimize radiotherapy regimens for patients with high-grade gliomas[33,34]. Another example includes the application of digital twin frameworks, personalized adaptive proton stereotactic body radiotherapy (SBRT) with efficient selection of a treatment plan selection and reduced target volume set-up uncertainty in patients with prostate cancer[35].

In this new paradigm, agentic AI can interact with the digital twin to test therapeutic strategies, optimize adaptive radiotherapy plans, predict adverse events and personalize clinical decision-making. Together, agentic AI and digital twins have the potential to enable proactive, continuous learning radiation oncology workflows, improve outcomes and inform the design of future clinical trials (Fig. 3).

AI applications across the radiation oncology workflow

AI has potential applications across the radiotherapy workflow, both to provide support for the automation of clinical tasks and as a predictive tool for clinical decision-making to deliver treatment in an adaptive manner (Fig. 1). Generative AI is generally used in supportive applications, whereas adaptive applications typically rely on predictive AI models. Automated approaches have enabled substantial gains in tasks such as organ segmentation, synthetic image generation, treatment planning, imaging guidance and quality assurance, thereby improving efficiency and consistency of care. Auto-segmentation has reduced contouring time from hours to minutes while achieving expert-level agreement for many disease sites[38–40]. AI-assisted treatment planning has improved consistency in plan quality and reduced planning time through automated optimization[41]. In image guidance, AI-based deformable image registration and synthetic computed tomography (CT) generation have facilitated adaptive radiotherapy by improving registration accuracy and enabling MRI-only treatment planning[42]. AI-driven quality assurance systems have further enhanced workflow efficiency by automatically detecting contouring, planning and delivery errors before treatment, reducing manual workload while maintaining patient safety. Together, these advances have improved efficiency, reproducibility and consistency across the radiation oncology workflow. Predictive models enable risk stratification, prediction of response to and toxicities from treatment, and adaptive treatment planning. These applications span the radiation oncology workflow, including automation of tasks (such as contouring, treatment planning, quality assurance and imaging guidance), analyses of response to treatment (personalized prediction of outcomes, re-irradiation and adaptative strategies), emerging multimodal approaches (for example, via integration of radiotherapy with systemic therapies such as immune checkpoint inhibitors in radioimmunotherapy) and, in the past few years, advances in generative AI and foundation models for treatment planning and delivery[43]. Advances include multimodal foundation models capable of jointly learning from radiological images, pathology slides, genomic data and clinical text to improve outcome prediction, and from clinical text to improve outcome prediction and clinical decision support. In parallel, LLMs have demonstrated promise for treatment planning assistance, contour review, treatment plan

documentation and retrieval-augmented clinical decision support[24]. Despite these advances, the current AI applications in radiation oncology remain largely task-specific and siloed, with limited interoperability across the workflow, highlighting the absence of a unified, continuously learning patient representation.

To address the fragmentation of task-specific AI tools, the concept of the digital twin has emerged as a framework for integrating patient-specific data across the radiotherapy workflow[44]. At present, however, the implementation of digital twins in radiation oncology remains incomplete and largely consists of disconnected, task-specific models spanning anatomy, dose, motion and treatment delivery, with limited integration of clinical, biological, outcomes and population-level data (Fig. 2). These 'partial' digital twins are often episodic, inconsistently updated and lack robust longitudinal integration across the care continuum[45] for scenarios such as clinical response, adaptive radiotherapy frameworks that update treatment plans using daily cone–beam CT or MRI-guided imaging, reinforcement learning models for sequential treatment adaptation and Bayesian optimization approaches for individualized dose scheduling. These generally operate as isolated applications focused on a single task and lack continuous bidirectional integration with the clinical workflow. Representative examples include patient-specific tumour growth models calibrated using serial imaging to simulate treatment. As a result, they fail to provide a persistent, continuously evolving representation of the patient, restricting their ability to support real-time adaptation, cross-domain reasoning and closed-loop decision-making — the latter concept referring to a radiotherapy framework in which planning, delivery, monitoring and adaptation are dynamically linked — across the course of treatment. This limitation highlights the considerable gap between current, partial digital twin implementations and the envisioned fully integrated, dynamically updated patient models. Unlike cardiovascular applications, such as coronary CT-derived fractional flow reserve, in which patient-specific computational models have been incorporated into clinical decision-making, most digital twin models in radiology, radiotherapy and theranostics remain at the framework developmental stage[46–48].

Synthetic imaging and multimodal AI. Synthetic imaging and data integration with multimodal AI are key areas of radiotherapy currently undergoing substantial advances. AI-driven synthesis of medical images has emerged as an essential component of clinical decision-making, particularly in settings requiring multiple imaging modalities[41]. Beyond generation of synthetic CT images, AI-based approaches have been developed for synthetic imaging across multiple modalities, including MRI and positron emission tomography, and for cross-modality synthesis translation between CT, MRI and positron emission tomography[49]. In particular, the generation of synthetic CT images from MRI data has received substantial attention for radiotherapy planning because it enables accurate dose calculation without the need for additional CT scans[50,51]. Clinical applications of MRI-derived synthetic CT have been most extensively investigated in prostate cancer and intracranial radiotherapy. More recently, studies have expanded to head and neck, and pelvic malignancies. These approaches have the potential to reduce patient burden and streamline workflows while maintaining dosimetric fidelity. AI-based MRI-to-CT synthesis demonstrated superior accuracy compared with conventional techniques such as atlas-based registration[52–54]. For example, a deep convolutional neural network (CNN) developed for patients with brain tumours reduced the mean absolute error from 94.5 ± 17.8 Hounsfield units (HU) with an atlas-based approach to 84.8 ± 17.3 HU (increased Pearson correlation coefficient 0.896–0.906)[55]. More broadly, cross-modality synthesis translation facilitates the integration of complementary imaging information for enhancing target delineation, treatment planning and response assessment, while reducing reliance on multiple imaging acquisitions.

These techniques address practical challenges associated with multimodal imaging acquisition, including cost, patient burden and image registration inaccuracies. By leveraging generative deep learning approaches, such as GANs and diffusion models, synthetic imaging approaches have demonstrated improved spatial accuracy and consistency compared with conventional approaches, although many applications remain investigational[50,54]. Impending challenges with

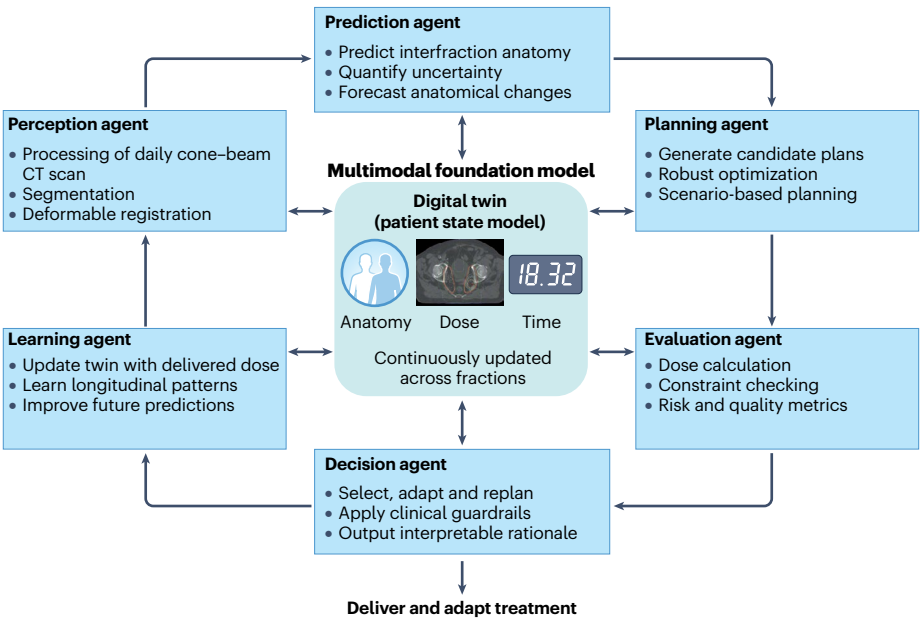


Fig. 3 | Proposed framework for the integration of digital twins, foundation models and agentic AI in radiotherapy. We propose a conceptual framework for closed-loop adaptive radiotherapy integrating digital twins, multimodal foundation models and agentic artificial intelligence (AI) systems. A digital twin constitutes a continuously updated patient-specific state model integrating longitudinal information across anatomy, dose, imaging and temporal changes throughout treatment. CT, computed tomography. Credit: Thomas Hecker/Alamy (Dose).

synthetic data remain; indeed, several studies have shown risks of missing relevant information known as 'mode collapse', leading to under-representation of less common clinical scenarios (such as small tumours, complex anatomy and artefacts)[56].

Automated segmentation and contouring. Automated segmentation and contouring of tumours and organs at risk constitutes one of the most mature and clinically adopted AI applications in radiotherapy. Deep learning architectures, particularly U-Net-based models, have demonstrated excellent accuracy in delineating anatomical structures across multiple disease locations, including the head and neck, thoracic, abdominal and pelvic regions. Contemporary deep learning auto-segmentation models have achieved Dice similarity coefficients frequently exceeding 0.85–0.90 for many organs of interest, approaching typical levels of interobserver agreement among expert clinicians[57].

This automation reduces interobserver variability and substantially decreases contouring time, thereby improving workflow efficiency[58]. For example, in a blinded randomized clinical validation study involving 42 organs at risk, deep learning-based auto-segmentation reduced total contouring time by 76% (3.4 h to 0.7 h) for head and neck organs at risk[59]. Moreover, in the setting of adaptive radiotherapy, rapid auto-contouring enables frequent and timely reoptimization of treatment plans on the basis of anatomical changes. Despite these advances, physician oversight remains essential to ensure clinical accuracy, and to account for complex or atypical tumours. As such, automated segmentation is best viewed as an assistive tool within a human-in-the-loop framework. Automated segmentation has become an essential component of contemporary radiation oncology workflows with several commercial tools that can offer high-quality contouring performance[60]. Nevertheless, additional work needs to be done to reliably and accurately segment gross tumour volumes in contrast to organs of interest. Even in such a case, performance can depend on the algorithm and its implementation. In a prospective randomized trial with results published in 2024, AI-assisted heart contouring did not reduce time compared with standard manual contouring, underscoring the need to evaluate AI tools for their intended use in real-world clinical workflows[61].

AI-driven treatment planning and dose optimization. Extending beyond segmentation, AI has also transformed treatment planning and dose optimization. Machine learning-based approaches, including KBP and deep learning models, enable the prediction of optimal dose distributions tailored to patient-specific anatomy. Conventional dose planning is an iterative and time-consuming process requiring repeated adjustments with the aim of optimizing and achieving target coverage while respecting dose to organs of interest. By contrast, AI-driven approaches can predict achievable dose distributions and streamline inverse planning, an automated process in which the prescribed target dose and organ constraints are predefined (as opposed to manually assigned and adjusted), reducing planning time while improving plan consistency. These models leverage prior high-quality plans to guide optimization and support reproducible, data-driven workflows. More-recent approaches leverage deep learning, CNNs and GANs to predict three-dimensional (3D) voxel-wise dose distributions. For example, a deep learning model based on the U-net architecture and its variants can be used for 3D voxel-wise dose prediction for irradiation of prostate[16] and head and neck[17] cancers. Conversely, early implementations of KBP focused on leveraging historical plan data to improve efficiency and consistency[62]. An evaluation of ten KBP models integrated within a fully automated treatment-planning workflow across ten distinct disease locations demonstrated that high-quality, clinically acceptable plans can be generated without routine manual optimization[63]. Upon review of the KBP-generated plans, physicians deemed that 88% of such plans were clinically acceptable without modification and 98% were acceptable after only minor edits, demonstrating the potential for robust automation across diverse disease locations[63]. Over the past few years, reinforcement learning has been applied to automate and improve inverse treatment-planning efficiency for brachytherapy[19] and external-beam radiotherapy[64,65]. Within these models, those using deep reinforcement learning (DRL) have shown improved accuracy and efficiency compared with KBP, although most AI treatment-planning models are largely investigational[66]. Direct comparative studies between DRL and KBP based treatment planning remain limited, although the currently existing studies suggest that AI-based planning approaches can achieve plan quality comparable to or, in select settings, exceeding KBP while reducing planning time[67]. Collectively, these advances have the potential to shift radiation oncology towards more personalized, adaptive and data-driven treatment paradigms.

From KBP to adaptive and agentic systems. Building on these advances, contemporary AI approaches are increasingly integrating KBP and DRL with adaptive radiotherapy, ablative SBRT, re-irradiation and radioimmunotherapy frameworks, among others. AI-enabled planning systems now leverage multimodal data and auto-dose prediction to streamline inverse planning and personalize optimization objectives on the basis of patient-specific characteristics[63]. In parallel, adaptive radiotherapy incorporates real-time imaging and predictive analytics to identify when treatment modification is required, maintaining optimal target coverage while minimizing toxicity.

Early clinical implementation of image-guided adaptive radiotherapy, such as MRI-guided adaptive workflows, has demonstrated the feasibility of modifying treatment plans in response to anatomical changes during the course of treatment[68]. Clinically, MRI-guided radiotherapy has provided a practical foundation for online adaptation: in a multi-institutional phase 2 trial testing MRI-guided on-table adaptive SBRT in 136 patients with borderline resectable and locally advanced pancreatic ductal adenocarcinoma, adaptive replanning was performed for the vast majority of delivered fractions (93%), demonstrating the feasibility of routine on-table adaptation at scale[68]. Additional prospective trials testing MRI-guided adaptive radiotherapy such as this one have demonstrated the feasibility of routine on-table adaptation while maintaining dosimetric quality and clinical workflow efficiency[69]. Furthermore, AI enhancement could support the transition from 'single-shot' planning to iterative decision support[66]. Together, these advances unify KBP, DRL-based planning and adaptive decision support into a cohesive framework that promises more efficient, data-driven and reproducible clinical workflows while preserving clinician oversight.

The evolution of AI-enabled radiotherapy extends beyond static plan generation towards adaptive, continuously learning treatment systems. Building on these foundations, AI methods increasingly aim to integrate real-time imaging, delivered dose reconstruction and predictive modelling into structured decision-support frameworks, therefore enabling a shift towards closed-loop radiotherapy frameworks. Agentic AI does not necessarily imply autonomous clinical decision-making, but rather systems capable of monitoring inputs, identifying predefined trigger thresholds and initiating transparent,

auditable decision-support actions within clinically supervised workflows. Together, these developments lay the groundwork for integrated AI and digital twin-enabled radiotherapy systems capable of continuous learning and personalized treatment adaptation.

Motion management and image-guided radiotherapy. Intrafraction motion remains a substantial challenge in radiotherapy, particularly for thoracic and abdominal tumours affected by respiration and organ deformation. Researchers have developed AI-based motion-tracking and image-analysis techniques to estimate real-time tumour position using fluoroscopy, cone–beam CT, MRI and surface imaging. These models enable dynamic tumour tracking, respiratory gating and adaptive beam delivery, improving targeting accuracy while reducing unnecessary irradiation of surrounding healthy tissue. Contemporary image-guided motion-management strategies can account for respiratory-induced tumour motion that might exceed 10 mm in thoracic and upper abdominal malignancies, enabling more precise target localization and narrower treatment margins[70]. In MRI-guided radiotherapy systems, AI-driven segmentation and motion modelling further support continuous visualization and online adaptation, improving the precision of image-guided treatment delivery. These capabilities are essential for enabling real-time feedback with closed-loop radiotherapy systems and form an essential component of adaptive and digital twin-driven treatment paradigms.

Outcome prediction, toxicity modelling and longitudinal follow-up. Predictive analytics is a rapidly expanding domain of AI in radiation oncology, with models developed to forecast response to treatment, toxicity to non-malignant tissues, disease control and survival outcomes. In the past few years, AI-based outcome-prediction models have demonstrated clinically meaningful predictive performance, with several studies reporting area under the curve values >0.80 for toxicity, disease recurrence and survival end points[71], although further validation is still warranted. Prediction models have traditionally relied on structured variables such as clinician-graded toxicities. However, the grading of adverse events outside of clinical trials is often inconsistently documented, and a large proportion of relevant information resides in unstructured clinical documentation and PROs. Advances in LLMs enable automated extraction and grading of adverse events from clinical notes. For example, in a benchmarking study with results published in 2025, LLMs showed high accuracy for binary calling of adverse effects and moderate agreement for grade-level classification, highlighting the potential of such models to enhance outcome capture at scale[72].

By leveraging multimodal inputs, including imaging features, dosimetric parameters, clinical variables and PROs, these models aim to support risk-adapted treatment strategies and personalized care[73]. For example, one model included 475 patients with early-stage non-small-cell lung cancer treated with SBRT and used longitudinally collected PROs together with clinical and demographic variables. The transformer model achieved a cross-validated area under the curve of 0.862 (c-index 0.753) for overall survival prediction and outperformed Cox and joint probability survival models[73]. AI-based models for the prediction of toxicities, for example, have demonstrated promise in identifying patients at elevated risk for radiation-induced complications, potentially enabling proactive mitigation strategies[74,75]. For example, a deep learning model developed with data from 784 patients with head and neck cancer enrolled in the RTOG 0522 trial integrated planning CT images, 3D dose distributions and salivary-gland contours to predict grade ≥2 xerostomia at 12 months after radiotherapy[75]. Longitudinal outcome modelling further enables continuous learning from real-world clinical data, providing a foundation for adaptive treatment approaches and the future development of dynamic AI–digital twin paired frameworks.

AI in patient education and clinical communication. LLMs have also been evaluated as tools for patient education in radiation oncology. In a prospective study, clinicians as well as patients with meningioma who had undergone radiotherapy highly rated LLM-generated responses to disease-related questions in terms of relevance, clarity and correctness[76]. Of note, patients expressed moderate-to-high trust in the information and willingness to use such tools for future education[76].

AI for handling clinical documentation and administrative burden. Ambient AI scribes are voice-enabled AI tools used in health care to generate clinical notes. Evaluations of these tools have demonstrated that the use of AI for documentation purposes can reduce the cognitive load on clinicians and improve workflow efficiency, reducing time spent on clinical documentation outside scheduled working hours by ~30% and improving self-reported clinician efficiency, cognitive workload and satisfaction[77,78]. Qualitative studies using semi-structured interviews across multiple specialties and practice settings confirm these observations[79]. However, clinical documentation continues to require physician review and editing to ensure accuracy[80]. Administrative processes such as prior authorization contribute to treatment delays. AI systems can assist by generating standardized, guideline-concordant prior authorization documentation, supporting medical necessity and streamlining administrative workflows. In this context, AI functions as a workforce multiplier rather than an autonomous decision-maker[81].

Machine-specific and patient-related quality assurance. Quality assurance frameworks for AI in radiation oncology have evolved across both machine-specific and patient-specific domains[82]. Several studies highlight that both conventional approaches and deep learning-based architectures (such as CNNs, recurrent neural networks and variational autoencoders) need to be integrated to predict beam characteristics, detect mechanical errors and estimate dosimetric differences in intensity-modulated radiotherapy and volumetric modulated arc therapy plans[83–85]. Specifically, gamma passing rate (GPR) is a quantitative dosimetric metric for patient-related quality assurance that enables comparison of the planned dose distribution and the measured delivered dose distribution. GPR is widely used to validate intensity-modulated radiotherapy and volumetric modulated arc therapy plans, detect multileaf collimation errors, assess delivery accuracy and compare predicted versus measured quality assurance; however, high GPR values do not guarantee the absence of clinically relevant errors. Reported mean absolute errors for GPR prediction fall within the acceptable threshold of 2–3%, suggesting that AI-based quality assurance approaches maintain safety standards while reducing measurement burden[86,87].

Outcomes modelling and survivorship care. Modelling of treatment outcomes is a key application of AI in radiation oncology[88]. Emerging models of survivorship and outcomes integrate multimodal data streams, including dosimetry, imaging radiomics and clinical variables, as well as longitudinal follow-up timepoints to dynamically estimate individual risk[89]. Meta-analyses suggests that hybrid models combining radiomics and clinical features outperform single-modality

approaches for predicting radiation-induced pneumonitis only[90] or simultaneously with local control[91] in patients with non-small-cell lung cancer. A systematic review and meta-analysis published in 2025 and including 56 studies comprising 12,803 patients with lung cancer demonstrated strong performance of machine learning models for predicting radiation-induced pneumonitis and immune checkpoint inhibitor-induced pneumonitis[90]. For radiation pneumonitis, pooled c-indices were 0.82 for dosiomics-based based models, 0.80 for radiomics-based models and 0.90 for models combining radiomics with clinical characteristics[90].

Clinical implementation

The implementation of AI in clinical practice involves multiple stages, including model validation, acceptance testing, commissioning, deployment, quality assurance, quality control and continuous performance monitoring. Despite rapid advances, challenges remain for the implementation of AI applications into routine workflows for radiation oncology; technical, operational and governance-related barriers include data heterogeneity, workflow integration and limited model generalizability across institutions[92,93].

Several frameworks have been proposed to guide the safe and effective integration of AI into radiation oncology, outlining structured pathways from model development to clinical deployment and monitoring[94,95]. These frameworks include recommendations for prospective validation, implementation and ongoing quality assurance within the radiotherapy workflow[96–99]. In parallel, regulatory oversight has a crucial role in ensuring that AI-enabled tools meet established safety and performance standards, with evolving frameworks and guidance from regulatory authorities in the USA, European Union (EU), UK and other regions.

Regulation of AI devices

The regulatory landscape for AI in health care is rapidly evolving, reflecting the need to balance innovation with patient safety and clinical accountability. In March 2024, the Food and Drug Administration (FDA) released updated guidance outlining key principles for the development and regulation of AI-enabled medical devices, particularly software as a medical device (SaMD)[100]. AI and machine learning-based software developed with the intention to treat, diagnose, cure, mitigate or prevent disease or other medical conditions is considered a medical device under the Federal Food, Drug, and Cosmetic Act and is commonly classified as SaMD by both the FDA and the International Medical Device Regulators Forum. The majority of these tools are cleared as Class II devices via the 510(k) pathway, which means that the product is 'substantially equivalent' to a legally marketed predicate device and is marketed for clinical use within specified indications. Machine learning-based SaMDs can be further adapted after deployment using real-world data, thereby enabling ongoing improvement of their performance. However, post-market changes can result in outputs different than those originally cleared for a set of inputs[100], highlighting the need for regulatory frameworks that accommodate controlled product evolution and a lifecycle-based approach to enable more personalized product customization services driven by customer demands (in this case, from health care) over the lifecycle of the product[101]. Despite progress in guidance development over the past few years, no federal legislation currently governs the use of AI in health care in the USA, which is largely decentralized and has a fragmented regulatory oversight. Instead, a combination of federal guidance and state-level initiatives has emerged. For example, the AI Act of May 2024 in Colorado (SB 24-205) addresses algorithmic accountability and patient-data privacy, whereas in Illinois, the AI Systems Act mandates coverage decisions for AI-driven health care. These initiatives demonstrate the growing recognition of the need for transparency, accountability and post-market surveillance.

In contrast to the FDA, the EU has adopted a more-centralized regulatory approach through the Medical Device Regulation (EU MDR) of 2017 and the EU Artificial Intelligence Act in 2024[102], which classify AI systems used with diagnostic, monitoring and clinical decision-support purposes as high-risk medical devices. This framework imposes strict requirements for validation, risk management, transparency and continuous post-market surveillance, with a strong emphasis on human oversight and auditability.

Professional societies also have an increasingly important role in defining best practices for the integration of AI into clinical workflows. In a joint guideline issued in June 2024, the American Association of Physicists in Medicine and the European Society for Radiotherapy and Oncology outline a comprehensive framework for the validation and clinical deployment of AI in radiotherapy[97]. They emphasize standardized validation, transparent reporting and model accessibility, reinforcing the importance of reproducibility and trust in clinicians.

Globally, regulatory approaches continue to vary. For example, China has adopted a regulatory framework emphasizing security, risk monitoring and assessment, and regulatory oversight of AI. The Chinese Cybersecurity Law was amended in 2025 to incorporate provisions addressing AI development, ethical standards, risk monitoring and assessment, and safety supervision, with the amendments taking effect on 1 January 2026. Personal information, health data and other data referred to as important that are collected or generated in China must generally be stored within China's borders. Moreover, post-market monitoring is government-led. The difference between this regulatory framework and those in other regions underscore the need for international harmonization of such frameworks, particularly as AI-enabled tools are deployed across diverse health-care systems.

Taken together, current regulatory efforts reflect an evolving balance between fostering innovation and ensuring patient safety. As AI systems become more adaptive and are increasingly integrated into clinical decision-making, future regulatory models will need to shift towards lifecycle-based oversight models that emphasize continuous validation, real-world performance monitoring and human-in-the-loop supervision. Within this regulatory context, FDA-cleared AI-enabled medical devices constitute the current standard for clinically deployable tools, providing important insights into how AI is being translated into real-world radiation oncology practice. Importantly, not all AI software developed for research purposes is approved by regulatory authorities. This distinction between regulatory approval and research-grade AI models underscores the importance of evaluating clinically validated tools. FDA-cleared platforms are clinically validated tools with established safety and performance profiles. As of May 2026, the FDA Artificial Intelligence-Enabled Medical Devices List included 31 unique AI-enabled tools spanning all major stages of the radiation oncology workflow (Table 1 and Supplementary Table 1). This list of FDA-cleared AI-enabled devices reflects the fact that the AI applications currently used in radiation oncology focus primarily on workflow support functions rather than autonomous clinical decision-making. Deep learning-based approaches predominate, accounting for 58% of the AI-enabled radiation oncology current applications, particularly for image segmentation and synthetic image generation. Although less commonly used, machine learning and hybrid models are the

Table 1 | Representative FDA-approved AI products across different steps of the radiation oncology workflow

Radiation oncology workflow step	Representative FDA-cleared AI tool	Primary function	Clinical relevance
Image synthesis and treatment planning preparation	MRI Planner (Spectronic Medical)	Deep-learning-based generation of synthetic CT images from MRI for MRI-only radiotherapy planning	Enables MRI-only treatment planning, reduces CT–MRI registration uncertainty and streamlines simulation workflows, synthetic CT generation can be completed in <5 min (ref. 126)
Contouring and segmentation	Siemens AI-Rad Companion	Automated segmentation of tumours and organs of interest	Reduces contouring time and interobserver variability while improving workflow efficiency; reported reductions in contour-editing time range from 15% to 86% depending on anatomical site[127,128]
Treatment planning and dose optimization	RayStation	AI-assisted treatment-planning dose optimization and planning automation	Improves planning efficiency, dose consistency and scalability of complex treatment planning; automated planning reduces average planning time by >43% compared with manual planning[128]
Image guidance and adaptive radiotherapy	Ethos	Cone–beam CT-guided adaptive radiotherapy with online replanning	Enables personalized treatment adaptation based on daily anatomical changes; in prostate radiotherapy, the adaptive plan was preferred in 95% of fractions and met more planning goals than the scheduled plan in 78% of them, and online recontouring and replanning can be completed in a mean of 19 min (ref. 129)
Response assessment and surveillance	HealthMyne	Radiomics-based quantitative imaging analysis and longitudinal monitoring	Facilitates assessment of response to treatment and disease surveillance; 80–84% concordance with radiologist RECIST classifications and interobserver agreement for quantitative volumetric delineation is >0.85 (ref. 130)

Products were selected as representative examples of Food and Drug Administration (FDA)-cleared artificial intelligence (AI)-enabled technologies spanning distinct stages of the radiation oncology workflow, with preference given to tools with published clinical validation data and/or evidence of clinical implementation (Supplementary Table 1). CT, computed tomography; MRI, magnetic resonance imaging; RECIST, Response Evaluation Criteria in Solid Tumors.

approaches favoured for treatment-planning support and workflow optimization (Supplementary Table 1). These trends suggest that the currently approved AI tools for radiation oncology focus on improving efficiency, standardization and reproducibility of clinical workflows while maintaining human oversight.

Clinical trials

Following the increasing number of FDA-cleared AI tools for radiation oncology, rigorous clinical evaluation to establish their safety, effectiveness and role in real-world clinical settings is a crucial need. Whereas regulatory approval reflects technical performance and safety within defined parameters, clinical trials are the gold standard for evaluating the safety and efficacy of new medical tools — including AI tools — and whether they meaningfully improve workflow efficiency, decision-making and patient outcomes.

As of May 2026, ClinicalTrials.gov listed 50 clinical trials evaluating AI applications across the radiation oncology workflow (Table 2 and Supplementary Table 2). Collectively, supportive and workflow-oriented applications — including contouring and segmentation, treatment planning, adaptive radiotherapy and image-processing workflows — accounted for ~60% of identified trials, whereas predictive applications accounted for ~40%. Among individual application categories, outcome prediction was the most common, constituting ~36% of trials. A smaller proportion of studies focus on outcome prediction, toxicity assessment, symptom monitoring or personalized treatment adaptation. In addition, the majority of identified studies are interventional feasibility or workflow-evaluation trials (66% versus 34% being observational studies), with fewer assessing the effect of AI on patient-centred end points (Supplementary Table 2).

Synthetic clinical trials

Generative AI models are being explored as a means to address insufficient patient accrual in oncology clinical trials. A validation study using data from nine completed clinical trials involving patients with breast cancer demonstrated that sequential synthesis models (a generative AI subtype) can simulate additional patients and replicate published treatment effects[103]. The investigators retrospectively simulated insufficient-accrual scenarios using these completed trial datasets to evaluate whether generative models could augment the missing patient population. Notably, the models reproduced treatment effects even when trained on datasets comprising ~60% of the original trial population, suggesting their potential to augment incomplete or underpowered trial datasets. Moreover, digital twins could have a relevant role in synthetic clinical trials[44,104]. In this context, virtual clinical trial validations of newly developed AI products could be conducted with digital twins and/or underpowered clinical trials could be augmented with accurate synthetic data to draw more meaningful conclusions about the performance of the AI tool being evaluated[34].

These applications highlight the broad potential of generative AI in radiation oncology; however, the clinical relevance of generative AI-based tools will ultimately depend on their ability to predict and optimize patient-specific outcomes. Outcomes modelling provides the unifying computational framework through which these diverse tools can be integrated into clinically actionable decision support (Fig. 3).

In a clinical decision-support computational framework, a continuously updated patient-specific digital twin integrates longitudinal imaging, delivered dose and temporal anatomical changes across treatment fractions[45,105,106]. A set of modular AI agents operates on this shared representation. For example, a perception agent processes daily cone–beam CT data to update anatomy using segmentation and deformable image registration. A prediction agent forecasts interfraction anatomical variations and associated uncertainties. A planning agent generates candidate treatment plans using robust, scenario-based optimization across predicted anatomies. An evaluation agent computes dose distributions, constraint adherence, and risk of irradiation of adjacent tissue and quality metrics on both the current and predicted anatomies. A decision and governance agent selects the optimal clinical action (selecting or adapting a plan, or replanning) under uncertainty-aware and clinically constrained conditions. Finally, a learning agent updates the digital twin after each fraction on the basis

Table 2 | Representative clinical trials evaluating AI applications across the radiation oncology workflow

Radiation oncology workflow step	Trial	Cancer type	Study type	Status	Primary end point
Contouring and segmentation	Use of AI-generated contours in radiation planning of the prostate brachytherapy (NCT06964412)	Prostate	Interventional	Active, not recruiting	Dice coefficient between final clinically approved brachytherapy prostate contours versus manual and AI-assisted contours provided by new learner
Treatment planning	Randomized evaluation of machine learning-assisted radiation versus standard treatment planning (NCT05979883)	Head and neck	Interventional	Completed, with no results available	Change in swallowing score
Adaptive radiotherapy	AI for head and neck cancer treated with adaptive radiotherapy (RadiomicART; NCT05081531)	Head and neck	Interventional	Active, not recruiting	Locoregional recurrence-free survival
Image guidance and response assessment	CT-guided stereotactic adaptive radiotherapy (CT-STAR; NCT05785845)	Lung	Interventional	Active, not recruiting	Incidence of grade ≥3 treatment-related toxicities
Prediction of clinical outcomes and treatment-related toxicity	Prediction of acute radiation-induced oral mucositis in nasopharyngeal carcinoma (NCT05858385)	Nasopharyngeal carcinoma	Observational	Recruiting/ enrolling	Incidence of severe radiation-induced oral mucositis

Trials were selected as representative examples spanning distinct artificial intelligence (AI) applications and stages of the radiation oncology workflow, with preference given to prospective and interventional studies evaluating clinically relevant measures of feasibility, performance, workflow integration or patient outcomes (Supplementary Table 2). CT, computed tomography.

of the observed anatomy and delivered dose, enabling continuous refinement over time. This closed-loop feedback architecture enables true longitudinal and adaptive radiotherapy, moving beyond static workflow-based approaches towards continuously learning, personalized treatment. Although fully integrated, continuously learning digital-twin systems remain in development, modelling studies have demonstrated the feasibility of individual components, including cone–beam CT-based anatomical updating, prediction of interfraction anatomical changes, automated treatment-plan generation and evaluation, and adaptive replanning[105].

Building on these advances, we expect the next generation of generative AI–digital twin paired systems to evolve into fully integrated, continuously learning and multimodal frameworks that support real-time clinical decision-making. This paradigm is centred on a persistent patient-specific digital twin that is continuously updated across treatment fractions and enriched through multimodal foundation models (Fig. 3). In this architecture, AI agents specialized in perception, prediction, planning, evaluation, decision-making and learning interact with the digital twin in a coordinated, closed-loop manner, enabling adaptive radiotherapy that dynamically responds to evolving patient anatomy, delivered dose and clinical context.

Framework for AI-based outcome modelling

In radiation oncology, outcome modelling involves mapping the correlations between high-dimensional patient-specific inputs and clinically relevant end points. Unlike traditional radiobiological models, which rely primarily on dosimetric parameters, contemporary approaches integrate heterogeneous data spanning clinical characteristics, treatment variables, imaging features and biological information. This shift reflects an evolving understanding that the response to radiotherapy is governed by complex, multiscale interactions that cannot be captured by simplified analytical formulations.

Contemporary AI-based outcome models are typically structured as modular, multimodal architectures in which each data modality is first encoded into a latent representation using modality-specific networks (such as CNNs for imaging or representation learning methods for biological data). For example, models developed in the past few years have integrated CT imaging, 3D dose distributions and clinical variables to predict treatment-related toxicity, while other multiomodal approaches have incorporated imaging, dosimetric and biological and/or clinical features for risk prediction. These studies demonstrate the current feasibility of integrating heterogeneous patient-specific and treatment-specific data within a unified predictive framework, although most of the current evidence remains retrospective[107,108]. These representations are subsequently fused into a shared feature space, enabling the joint prediction of multiple clinically relevant end points, including tumour control, treatment-related toxicity, survival and PROs. Importantly, the ability to model competing or complementary end points within a unified framework is a considerable advance over traditional approaches that treat these outcomes independently.

From a clinical perspective, outcome models support three principal functions in radiation oncology: (1) treatment planning by enabling optimization of tumour control while minimizing side effects; (2) personalization through individualized risk estimation; and (3) adaptation by updating predictions over the course of treatment. However, despite promising performance in retrospective studies, the implementation of these models in clinical practice remains limited by challenges related to data heterogeneity, generalizability and lack of prospective validation.

To address these limitations, a rigorous development and validation pipeline is required, encompassing standardized data curation, robust training and fine-tuning strategies, and both internal and external validation. In addition, consideration of interpretability, uncertainty estimation and algorithmic bias is key to ensure safe and equitable deployment. Looking forward, the integration of longitudinal data, causal modelling and real-time learning systems is expected to transform outcome modelling from a static predictive tool into a dynamic component of adaptive, personalized radiotherapy, forming the computational foundation for digital twin-enabled clinical decision-making.

Step 1: problem definition and end point specification. The first step in outcome model development is to clearly define the clinical question,

including the selection of relevant end points and patient-specific variables. End points might include tumour control outcomes (for example, local control or recurrence) and safety outcomes (for example, pneumonitis or xerostomia) as well as survival, and should be chosen on the basis of clinical relevance and measurability. The modelling paradigm (classification versus time-to-event analysis) and required multimodal inputs (clinical, treatment, dosimetric, imaging and/or biological) must be established at this stage.

Stage 2: data curation and harmonization, and cohort definition. Patient cohorts are defined on the basis of treatment characteristics and follow-up, with careful attention to inclusion and exclusion criteria. Data are curated to ensure quality, completeness and representativeness, including handling of missing data, standardization (using parameters such as *Z*-score or methods such as min–max normalization), harmonization across sources (for example, Common Terminology Criteria for Adverse Events versions) and mitigation of bias in under-represented populations. Nevertheless, using uncurated real-world datasets when developing large-scale foundation models using techniques such as self-supervised and/or weak-supervised learning remains a valuable approach, as discussed earlier.

Step 3: model development and optimization. Datasets are partitioned into training, validation and test sets to mitigate data leakage and minimize overfitting. Cross-validation strategies and class-balancing techniques (such as the synthetic minority over-sampling technique) are applied within the training set. Feature engineering or automated representation learning is performed, and models are developed with optimization of architecture and hyperparameters using validation data.

Step 4: validation, evaluation and interpretability. Model performance is rigorously evaluated using internal and external validation cohorts, with emphasis on generalizability, calibration and reproducibility. Discrimination, calibration and clinical utility metrics are assessed across clinically relevant subgroups. Importantly, this evaluation must extend beyond aggregate performance to include robustness to distributional shift, with explicit assessment of out-of-bounds or out-of-distribution inputs relative to the training data. Models should be tested for their ability to detect and appropriately respond to such inputs, including uncertainty estimation or abstention mechanisms (that is, declining to make a prediction) when predictions might be unreliable. Interpretability approaches[109] (including model-specific methods, model-agnostic methods such as local interpretable model-agnostic explanations (LIME) or Shapley additive explanations (SHAP), or saliency mapping) are incorporated to support clinical transparency, facilitate human–AI interaction and ensure that model behaviour aligns with clinical reasoning.

Step 5: bias assessment, safety and clinical integration. Systematic bias and fairness are evaluated across demographic and clinical subgroups[110]. Safeguards such as out-of-distribution detection and uncertainty estimation are incorporated to prevent inappropriate use of the model. Models that meet these criteria are integrated into clinical workflows with ongoing monitoring, regulatory oversight and human-in-the-loop decision-making to ensure safe and effective deployment.

Current challenges and recommendations

The clinical implementation of AI in radiation oncology is accompanied by a broad range of technical, methodological, ethical and operational challenges, which span infrastructure limitations, data quality and accessibility, model development and validation, regulatory oversight and workforce readiness. Addressing these barriers is essential to ensure safe, effective and equitable integration of AI into clinical workflows (Table 3). Explainable AI approaches can provide clinicians with interpretable information about the factors contributing to model outputs, supporting evaluation of AI-assisted decisions[109,111]. Taken together, these challenges are not theoretical, but instead are increasingly being observed in early clinical deployments of AI-enabled contouring, adaptive radiotherapy and outcome prediction systems in radiation oncology.

Technical requirements

In terms of software and hardware requirements, it is important to distinguish between the development and deployment phases. Model development often relies on high-performance computing and parallel processing, whereas clinical deployment prioritizes efficiency, reliability and real-time usability. For example, auto-contouring systems that require substantial manual correction can paradoxically increase clinician workload despite high segmentation accuracy. To address these challenges, AI tools should be designed with workflow integration as a primary objective, leveraging vendor-neutral architectures and standardized interfaces, although proprietary vendor-provided ecosystems that integrate AI solutions might need to be prioritized, particularly in smaller non-academic centres. Prospective evaluation within clinical workflows is essential to ensure usability and minimize disruption.

Data engineering

The performance and safety of AI models in radiation oncology are fundamentally determined by the quality, representativeness and structure of training data. Radiotherapy datasets often vary across institutions owing to differences in imaging protocols, contouring conventions, dose-prescription practices and toxicity documentation. The resulting heterogeneity can negatively affect model generalizability. A key challenge is that training datasets might not adequately represent the diversity of clinical practice, resulting in biased or non-generalizable models. In radiation oncology, the use of non-representative models can manifest as reduced performance of segmentation or toxicity-prediction models when applied to under-represented patient subgroups, tumour locations or institutions. To address these limitations, efforts have focused on establishing standardized data ontologies for structures and outcomes, rigorous data curation pipelines, and multicentre data-sharing initiatives[112,113]. Federated learning has also emerged as a promising approach, which enables distributing model training across institutions without direct data sharing, thereby improving scalability while preserving patient privacy[114–116].

Model development validation

Robust validation strategies are essential to ensure safe implementation of AI models into radiation oncology practice. Many models demonstrate robust internal performance; however, external validation frequently reveals reduced accuracy owing to dataset shift, including differences in imaging protocols, contouring practices and patient populations across institutions. Examples include toxicity-prediction models and automated segmentation systems[117–119]. As a result, multicentre validation, calibration testing and subgroup analyses are key components of model evaluation. Furthermore, prospective studies embedded within clinical workflows are needed to assess not only

Table 3 | Key challenges and recommendations for the implementation of AI in radiation oncology workflows

Domain	Key challenges	Practical recommendations	Representative examples in radiation oncology
Technical integration	Limited interoperability with treatment-planning systems, oncology information systems[a] and imaging platforms; workflow disruption	Develop vendor-neutral interfaces; integrate AI into existing workflows; test workflows prospectively	Auto-contouring tools not seamlessly integrating into treatment-planning systems[129]; commercial platforms requiring manual export and import workflows[131]
Data quality and curation	Heterogeneous, incomplete and biased datasets; lack of standardized annotations; labelling burden	Standardize data ontologies; implement QA pipelines; curate multicentre datasets	Variability in contouring labels[112]; inconsistent documentation of treatment-related toxicities across institutions[132]
Model development and validation	Overfitting, dataset shift; limited external validation; poor generalizability	Perform multicentre external validation; calibration testing; subgroup analysis; continuous performance monitoring	Models for prediction of treatment-related toxicities failing to generalize across institutions and having reduced performance on external validation[133]
Clinical utility and effect in workflows	Performance metrics (such as Dice score, AUC and F1 score) might not translate to clinical benefit; unclear gains in efficiency	Conduct prospective clinical trials; evaluate workflow efficiency and clinical end points; clinician-centred design	Geometric segmentation performance metrics may not translate into proportional reductions in clinical editing time across structures and institutions[134–136]
Ethics and bias	Algorithmic bias; unequal performance across populations; lack of transparency	Perform bias audits; ensure datasets are representative; improve model interpretability	Reduced model performance in under-represented patient populations[137,138]
Regulatory and governance	Evolving regulatory frameworks; challenges with adaptive or continuously learning models; liability concerns	Establish governance structures; ensure auditability; perform local validation before deployment	FDA-cleared tools requiring institutional validation before clinical use[139]
Safety and human oversight	Automation bias; risk of silent model failure; over-reliance on AI outputs	Maintain human-in-the-loop review; implement safety checkpoints and fail-safe mechanisms	Contouring errors persisting into treatment planning despite clinician review[140]
Workforce training and adoption	Limited AI literacy; resistance to workflow change; lack of structured training	Develop multidisciplinary training programmes; define roles; integrate AI education into clinical practice	Need for training of physicians, physicists, dosimetrists and therapists[123]

[a]Oncology information systems are an oncology-specific type of electronic health record. AUC, area under the curve; AI, artificial intelligence; FDA, Food and Drug Administration; QA, quality assurance.

predictive performance, but also clinically relevant end points such as workflow efficiency, treatment quality and patient outcomes.

Ethics and regulations

In radiation oncology, AI systems introduce important ethical and regulatory considerations, particularly given their role in high-stakes decision-making such as target delineation and dose optimization. Regulatory agencies, such as the FDA, have a central role in evaluating the safety and effectiveness of AI-based medical devices; however, regulatory clearance alone does not guarantee consistent performance across clinical environments. In radiation oncology, in which workflows vary considerably across institutions, additional local validation is often required before deployment. Moreover, adaptive or continuously learning AI systems introduce new challenges for regulatory oversight, including version control, transparency and post-deployment monitoring. These issues highlight the need for evolving regulatory frameworks that can accommodate dynamic AI systems used in clinical practice.

The FDA's Predetermined Change Control Plan framework for AI-enabled device software functions is an evolving approach that allows manufacturers to prospectively specify anticipated model modifications, methods for their development and validation, and approaches for assessing their effect[120]. This framework is intended to enable controlled iterative modification of AI-enabled devices while maintaining reasonable assurance of safety and effectiveness throughout the lifecycle of the product. Complementary Good Machine Learning Practice principles further emphasize representative datasets, independent testing, human–AI team performance and monitoring of deployed models across the total lifecycle of the product[121].

Governance

Governance systems create a legal framework to monitor and ensure safe and responsible implementation of AI in radiation oncology. As the role of AI continues to grow, the responsibility for AI-assisted decision-making becomes increasingly distributed across multiple stakeholders, including clinicians, physicists, administrators and vendors. A report on the governance structure of a single institution in oncology (not restricted to radiation oncology only)[122] identified the following minimum components: (1) presence of a multidisciplinary governance committee; (2) defining the intake process for new AI model ideas; (3) a risk-assessment framework; (4) a model registry; (5) access to vendor monitoring data; (6) linkage into the existing reporting system for safety events in the hospital; and (7) institutional support.

Human in the loop

The role of autonomous AI applications is intriguing because of its high efficiency; however, its use in health-care applications is too risky, and regulating human–AI interaction is necessary for success. In this context and despite advances in automation, radiation oncology remains a safety-critical domain. Human oversight is therefore essential to ensure clinical safety and maintain accountability. AI-generated outputs, such as auto-segmented structures or dose predictions, must be reviewed and validated by clinicians before use in treatment planning. Expert oversight is particularly important in situations in which errors can propagate downstream, such as incorrect organ-of-interest contours influencing inverse planning optimization and resulting in unintended dose distributions. Risks such as automation bias and silent model failure further reinforce the need for structured human-in-the-loop workflows. Clearly defined validation checkpoints, escalation pathways and audit

mechanisms are essential to ensure that AI serves as a decision-support tool rather than an autonomous decision-maker. Maintaining human oversight is crucial for preserving clinical accountability and trust. For high-risk clinical applications, human oversight should extend throughout the radiotherapy workflow, rather than being limited to final approval alone. For example, in an AI-assisted treatment-planning workflow, clinicians should review auto-segmented targets and organs at risk before these structures are used for dose optimization, rather than relying solely on review of the final treatment plan.

Workforce training

Finally, the successful implementation of AI in radiation oncology will depend on workforce readiness across all professional roles. Clinicians, medical physicists, dosimetrists and radiotherapists must develop their foundational literacy in AI tools, understanding the capabilities, limitations and potential failure modes of these systems. Structured multidisciplinary training programmes are needed to support safe adoption, particularly as AI becomes increasingly embedded into radiotherapy workflows. Clear definition of roles and responsibilities will be essential to ensure appropriate use, oversight and interpretation of AI outputs across the radiotherapy workflow. Professional societies have begun addressing this need through dedicated educational initiatives, including AI-focused continuing education through efforts from the American Society for Radiation Oncology (ASTRO) and American Association of Physicists in Medicine (AAPM) to develop formal curricula in information technology, data science and AI for radiation oncology professionals[123–125].

Conclusions

AI is rapidly transforming the landscape of radiation oncology, with applications spanning the entire clinical workflow from image segmentation to outcomes prediction and post-treatment monitoring. AI-enabled applications are moving radiation oncology from static treatment paradigms towards data-driven and adaptive clinical paradigms. By integrating automation, predictive modelling and multimodal data, AI supports more efficient workflows while enabling increasingly personalized treatment strategies. These advances lay the foundation for integrated digital twin frameworks that continuously update patient-specific anatomical, biological and outcome data. Building on this foundation, emerging agentic AI systems introduce a higher level of orchestration by enabling coordinated, context-aware decision support across the radiotherapy workflow. Rather than replacing clinician judgement, these systems continuously monitor inputs, identify predefined triggers and facilitate transparent, auditable decision-making within a human-in-the-loop[111] framework.

This Review is not intended to be exhaustive nor to cover every aspect of AI that will affect radiation oncology but, rather, to provide examples of successful AI applications. These examples demonstrate that, in the mid-term, AI-enabled tools will be integrated into clinical practice primarily to improve workflow efficiency, supported by a

Glossary

Agentic AI
AI systems composed of multiple agents that are orchestrated autonomously or semi-autonomously to execute complex tasks by breaking them down into smaller interdependent steps carried out by these agents.

Automation bias
The tendency for users to over-rely on recommendations or outputs generated by automated systems, potentially overlooking contradictory clinical information or errors.

Continuous learning
The process of periodically or continuously updating an AI model using newly available data to maintain or improve performance as clinical practices, patient populations or data distributions evolve.

Data leakage
The inadvertent use of information from the validation or test data during model training, resulting in overly optimistic estimates of model performance and reduced generalizability to new data.

Deep learning
Subfield of machine learning that uses neural networks to learn directly from raw data.

Explainable AI
AI methods that make model outputs understandable to humans, supporting transparency and trust in decision-making.

Fine-tuning strategies
In the context of AI, process of adapting a pretrained model to a specific task or dataset through additional training on domain-specific data.

Foundation model
Large-scale model trained on broad, diverse datasets that can be adapted to a wide range of downstream tasks (such as segmentation, planning or outcome prediction) with minimal task-specific training.

Generative AI
Models that can generate novel content (text, images, code, audio and so on) by learning patterns from large datasets.

Hallucination
In the context of AI, this term refers to a model generating incorrect or fabricated outputs that are not grounded in the input data or in reality.

Large language model
(LLM). A type of generative AI trained on very large text corpora using self-supervised learning that can understand and generate human-like language.

Machine learning
Subfield of AI that uses computer algorithms that can learn from previous experiences without being explicitly programmed for every rule.

Multimodal AI
AI model that integrates and processes multiple data types (such as imaging, text, genomics and clinical data) to generate unified predictions that, in radiation oncology, might be relevant to agentic orchestration and adaptive workflows.

Reinforcement learning
Machine learning paradigm in which models 'learn' to make decisions by optimizing actions on the basis of feedback from an environment.

Silent model failure
Deterioration or failure in AI model performance that occurs without an obvious warning to the user, potentially resulting from factors such as data or distribution shift.

Uncertainty estimation
Methods used to quantify the confidence of model predictions, often used to flag low-confidence outputs requiring human review.

Vision–language model
(VLM). A type of generative AI (multimodal foundation models) trained on very large multimodal data (such as text, images or videos corpora) that can generate visual data in response to such inputs.

Box 1 | Dos and don'ts of AI in radiotherapy

Dos

- Train artificial intelligence (AI) models on dataset clinically representative for a particular workflow (such as computed tomography simulation protocols, contouring standards or intensity-modulated radiotherapy or volumetric modulated arc therapy planning).
- Use rigorously validated AI tools to support clinical decisions within defined workflow steps (for example, auto-contouring, plan optimization or image guidance).
- Benchmark outputs against clinically validated reference ('golden standard' data) and predefined performance criteria appropriate to the intended task, such as contour accuracy, dose–volume constraints and plan for quality metrics, and integrate task-specific quality assurance checks within the clinical workflow before deployment.
- Continuously monitor AI performance after deployment for model drift and changes in task-specific performance, including contouring accuracy and plan quality, with recalibration or revalidation when indicated.
- Identify and evaluate failure modes using principles of failure modes using established risk-analyses methods, including failure mode and effect analysis and fault-free analysis[141].
- Document AI-assisted decisions in the treatment record, including model version, inputs and any clinical edits or overrides.

Don'ts

- Do not deploy models using data exposed during training (or 'data leakage').
- Do not deploy models trained in populations or clinical settings that are not adequately represented in the training and validation data (for example, single-institution contouring data).
- Do not rely solely on an AI output without oversight from a clinical provider, and do not accept auto-contours or treatment plans without expert review.
- Do not ignore model limitations, bias or variability across disease locations, and fractionation schemes, and do not overlook failure models in settings such as unusual anatomy, implants, prior irradiation or poor image quality.
- Do not use AI tools outside their intended indications (or 'off-label' use) or validated scope (for example, applying a model developed for prostate cancer both in the postoperative and re-irradiation settings).

growing body of FDA-cleared technologies and an expanding portfolio of prospective clinical trials. We all must recognize that AI is a rapidly evolving field with new FDA devices cleared, clinical trials initiated and technologies emerging rapidly; therefore, although we believe that the lessons drawn from this current broad overview will be applicable in the foreseeable future, we also expect that these changes will further transform the practice of radiation oncology as we know it today, with a newer set of opportunities as well as challenges not covered in this Review.

Despite remarkable clinical progress, the clinical implementation of AI tools remains constrained by important challenges including limitations in data quality, external validation and generalizability, as well as regulatory considerations and barriers to broad adoption by the workforce. Addressing these challenges will require coordinated efforts across clinicians, physicists, industry and regulatory bodies, as well as the continued development of standardized frameworks. The future of AI in radiation oncology lies in 'augmented intelligence', in which human expertise remains central. Successful integration of AI into radiation oncology will depend on the development of interoperable ecosystems that integrate digital twins and agentic frameworks (Fig. 3). With thoughtful implementation, AI has the potential to fundamentally enhance the precision, personalization and quality of cancer care. Crucially, these systems must be deployed within transparent, auditable and human-in-the-loop frameworks to ensure safety, trust and clinical accountability (Box 1).

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Competing interests
E.K. owns stocks in Abbvie, Bristol-Myers Squibb and Pfizer, and is one of the founding members of Digital Twin for Health. I.E.N. acts as co-Editor-in-Chief of *BJR Artificial Intelligence* and deputy editor of *Medical Physics*, and is a co-founder of iRAI Technologies.

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